

Experiment :- 20

Problem statement:-

Mathematical computations such as finding the Greatest Common Divisor (GCD), Least Common Multiple (LCM), and checking whether a number is prime are fundamental problems in programming and algorithm design. Automating these computations using shell scripting helps students understand arithmetic operations, conditional logic, and iterative constructs in Bash.

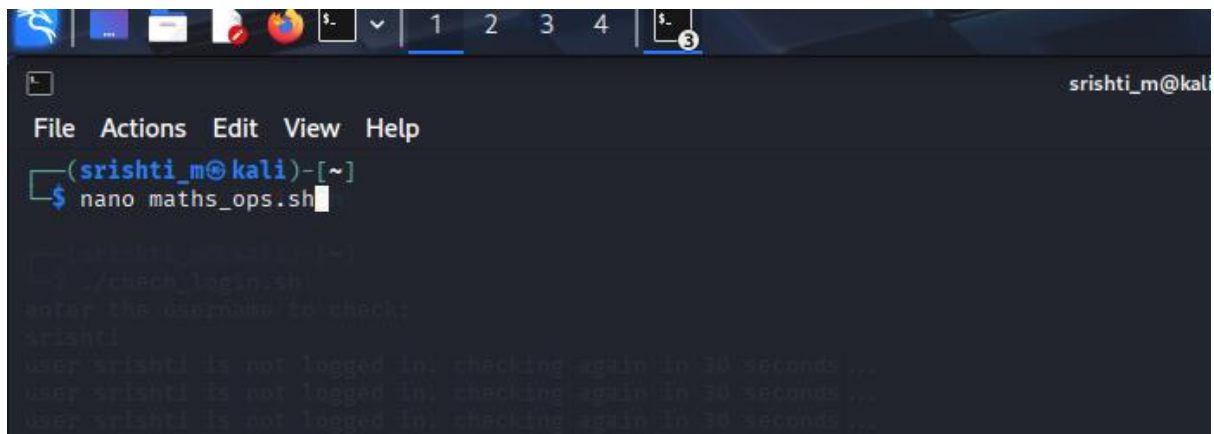
The problem is to write a shell script that accepts user input to compute the GCD and LCM of two numbers and checks whether a given number is prime, demonstrating arithmetic operations, loops, and decision-making in Bash scripting.

Aim:- Shell script to compute GCD. LCM and check whether a given number is prime.

Procedure:-

A. Writing the Shell Script

1. Open the terminal.
2. Create a script file: nano math_ops.sh



```
(srishti_m@kali)-[~]
$ nano math_ops.sh

(srishti_m@kali)-[~]
$ ./check_login.sh
enter the username to check:
srishti
user srishti is not logged in, checking again in 30 seconds ...
user srishti is not logged in, checking again in 30 seconds ...
user srishti is not logged in, checking again in 30 seconds ...
```

3. Enter the script:

```
File Actions Edit View Help
GNU nano 8.4 maths_ops
#!/bin/bash
echo "enter first number:"
read a
echo "enter second number:"
read b
# GCD calculation
while [ $y -ne 0 ]
do
    temp=$y
    y=$((x % y))
    x=$temp
done
gcd=$x
lcm=$(( (a*b) / gcd ))
echo "GCD of $a and $b is: $gcd"
echo "LCM of $a and $b is: $lcm"
echo "enter a number to check for prime :"
read n
flag=0
if [ $n -le 1 ]; then
    flag=1
fi
for ((i=2; i≤n/2; i++))
do
    if [ $((n % i)) -eq 0 ]; then
        flag=1
        break
    fi
done
if [ $flag -eq 0 ]; then
    echo "$n is a prime number"
else
    echo "$n is not a prime number"
fi
```

```
File Actions Edit View Help
GNU nano 8.4
# GCD calculation
while [ $y -ne 0 ]
do
    temp=$y
    y=$((x % y))
    x=$temp
done
gcd=$x
lcm=$(( (a*b) / gcd ))
echo "GCD of $a and $b is: $gcd"
echo "LCM of $a and $b is: $lcm"
echo "enter a number to check for prime :"
read n
flag=0
if [ $n -le 1 ]; then
    flag=1
fi
for ((i=2; i≤n/2; i++))
do
    if [ $((n % i)) -eq 0 ]; then
        flag=1
        break
    fi
done
if [ $flag -eq 0 ]; then
    echo "$n is a prime number"
else
    echo "$n is not a prime number"
fi
```

4. Save and exit the editor.

5. Make the script executable: `chmod +x math_ops.sh`

```
srishti_m@kali: ~  
File Actions Edit View Help  
(srishti_m@kali)-[~]  
$ nano maths_ops.sh  
(srishti_m@kali)-[~]  
$ chmod +x maths_ops.sh  
enter the username to check:  
(srishti_m@kali)-[~]  
$  
user srishti is not logged in, checking again in 30 seconds...  
user srishti is not logged in, checking again in 30 seconds...  
user srishti is not logged in, checking again in 30 seconds...  
user srishti is not logged in, checking again in 30 seconds...  
user srishti is not logged in, checking again in 30 seconds...  
user srishti is not logged in, checking again in 30 seconds...  
user srishti is not logged in, checking again in 30 seconds...
```

B. Executing the Script 6. Run the script:

7. `./math_ops.sh`

```
srishti_m@kali: ~  
File Actions Edit View Help  
(srishti_m@kali)-[~]  
$ nano maths_ops.sh  
(srishti_m@kali)-[~]  
$ chmod +x maths_ops.sh  
enter the username to check:  
(srishti_m@kali)-[~]  
$ ./maths_ops.sh logged in, checking again in 30 seconds...  
enter first number: logged in, checking again in 30 seconds...  
5 logged in, checking again in 30 seconds...  
enter second number: logged in, checking again in 30 seconds...  
3 logged in, checking again in 30 seconds...  
./maths_ops.sh: line 16: lcm=15: command not found seconds...  
GCD of 5 and 3 is: 1 logged in, checking again in 30 seconds...  
LCM of 5 and 3 is: cm logged in, checking again in 30 seconds...  
enter a number to check for prime : logged in, checking again in 30 seconds...
```